Machine Learning Model for Predicting the Risk of Childhood Stunting in Peru

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Abstract

Objective: To develop and evaluate a model that predicts the risk of stunting in children under five years of age in Peru, where prevalence has remained around 12% since 2018, and to implement it in a prioritization tool.

Methods: We analyzed microdata from the 2007–2025 Demographic and Family Health Survey (304,271 children) following the CRISP-DM framework. After a four-stage feature selection (45 variables), six algorithms were compared using stratified five-fold cross-validation, with class weights and the survey expansion factor. Uncertainty was estimated by bootstrap, and associated factors were interpreted with SHAP values. A panel of four health professionals evaluated the application with an instrument based on TAM2.

Results: LightGBM achieved the best performance (AUC-ROC 0.83; 95% CI: 0.827–0.832), with a sensitivity of 76.4% and an F1 score of 49.3%; with a threshold of 0.45, it detected 80.5% of cases. Household poverty, birth weight, size at birth and altitude were the factors most associated with risk, with different profiles across departments. All panel constructs reached a median of 4 or higher out of 5.

Conclusions: Routine survey data make it possible to prioritize the children at highest risk of stunting and to explain each result. The tool supports targeting and triage, not diagnosis.

Keywords: Stunting, Machine learning, LightGBM, SHAP, Peru, Triage

1. Introduction

Childhood stunting is one of the main public health challenges worldwide, as it affects the physical, cognitive and social development of children under five years of age [1, 2]. Its effects persist into adulthood: it is associated with shorter height, fewer years of schooling and lower economic productivity [3], and each improvement in height-for-age is related to better cognitive development [4]. Growth faltering begins in the first months of life and is rarely reversed [5], which is why the World Health Organization set the target of reducing the number of stunted children by 40% by 2025 [6]. Stunting is a complex phenomenon, influenced by economic conditions, maternal education, access to basic services and territorial inequalities [1]; although its prevalence is declining in low- and middle-income countries, socioeconomic gaps persist [7].

In Peru, stunting declined steadily during the 2000s [8], owing to improvements in parental education, maternal and neonatal care, and migration to cities [9]. However, the reduction was slower in rural areas [10], and in the Demographic and Family Health Survey (ENDES) the national prevalence has remained around 12% since 2018, with a persistent gap between rural and urban areas (Section 3.1). This stagnation suggests that general strategies are no longer sufficient and that resources should be directed to the children and territories at highest risk, in line with the evidence on scaling up proven nutrition interventions [11, 12]. In the country, malnutrition in all its forms remains concentrated in the poorest households [13].

Over the last decade, several studies have applied machine learning to predict child undernutrition with data from the Demographic and Health Surveys (DHS). In Bangladesh, Ethiopia and Zambia, tree ensembles obtained the best results [14, 15, 16], and other studies with DHS data identified the determinants of undernutrition using similar classifiers [17, 18]. In Rwanda, a neural network achieved an AUC-ROC of 0.84 [19]. In *Nutrition*, recent studies combined several algorithms into an ensemble for fourteen sub-Saharan African countries [20], identified the risk factors for wasting in Egypt [21], integrated satellite data to estimate household stunting in India [22] and predicted undernutrition among schoolchildren in the Philippines [23]. Other studies incorporated explainability techniques to identify the determinants of risk [24] or class-balancing techniques with accuracies close to 100% [25].

Despite these advances, limitations remain. A meta-analysis of studies with DHS data found a pooled accuracy of 68.9% for stunting and very high heterogeneity between studies [26]. Most studies rely on a single survey round, report mainly accuracy, a metric of little use when cases are a minority, and do not offer a tool that health professionals can use. In addition, these studies have been carried out mostly in Asia and Africa, and in Peru no predictive approach with territorial explanation has been applied to the full ENDES series.

Therefore, the aim of this study is to develop a machine learning model that predicts the risk of childhood stunting in Peru, makes it possible to analyze the predominant factors by department, and estimates nutritional risk at the individual and territorial levels.

This study is justified on three complementary grounds. On theoretical grounds, it incorporates social, environmental and territorial determinants into the analysis of child undernutrition through machine learning, beyond statistical models centered on anthropometric variables. On practical grounds, it offers a tool that identifies the children at highest risk and the predominant factors in each department, so that institutions such as the Ministry of Health, the Ministry of Development and Social Inclusion and regional governments can prioritize their interventions. On methodological grounds, it proposes a replicable workflow on a public survey spanning nineteen years.

The contributions of this study are as follows: 1) the cleaning and structuring of a consistent database that integrates national child nutritional status from the 2007–2025 ENDES microdata; 2) the systematic comparison of machine learning algorithms (Logistic Regression, Decision Tree, XGBoost, LightGBM, CatBoost and Neural Network) under the same stratified cross-validation protocol, in order to select the model with the best predictive performance; 3) the evaluation of the model with metrics suited to imbalanced classes (sensitivity, precision, specificity, F1, AUC-ROC and AUC-PR) and the interpretation of the influence of the variables through SHAP values, which increases its explanatory capacity; and 4) the implementation of an interactive territorial analysis system that shows the risk of stunting for the profile entered, making it easier for public health institutions to apply it.

This article is organized as follows: Section 2 describes the methodology, including the CRISP-DM framework, the construction of the database, the modeling and the application developed. Section 3 presents the results of the algorithm comparison and the interpretation of variables with SHAP. Section 4 discusses the interpretation of the results, their comparison with similar studies and the limitations of the study. Finally, Section 5 presents the conclusions, restating the aims, the main findings and the lines of future work.

2. Methods

The study followed the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework [27] (Figure 1). Business understanding is presented in the Introduction; this section describes the other five phases: data understanding, data preparation, modeling, evaluation and deployment. In these phases, the ENDES microdata were integrated, the predictor variables were selected, six algorithms were compared, the chosen model was evaluated and interpreted, and an application was implemented, whose relevance was evaluated by health professionals.

2.1. Data understanding

The study used the annual ENDES microdata, collected and published by the National Institute of Statistics and Informatics (INEI) of Peru between 2007 and 2025 and available on the INEI microdata portal (`https://proyectos.inei.gob.pe/microdatos/`). To build the unit of analysis, an extraction and relational integration workflow was designed that linked five modules through the keys `HHID` and `CASEID`. Module RECH0 provided the interview metadata; RECH1, the demographic characteristics of household members; RECH6, the anthropometric and child health indicators; RECH23, housing and sanitation conditions, as well as geographic altitude, which is related to hemoglobin concentration in the Peruvian child population [28]; and REC41, maternal obstetric history, prenatal care and birth weight. The latter was linked one-to-one with the child record through the bridge table REC21. The integration and the child–mother consistency audit showed a match above 99.9% in sex and year of birth. The consolidated sample included 304,271 children under five years of age with valid anthropometric records; Figure 2 summarizes the integration structure.

From this sample, the dependent variable, childhood stunting, was defined according to the World Health Organization growth standards [29], a criterion used consistently in the analysis of child undernutrition in Peru [8, 10]. A positive case ($Y = 1$) was defined as any child whose height-for-age Z-score (HAZ = HC70/100) was below minus two standard deviations from the reference median, and a negative case ($Y = 0$) otherwise (Equation 1). Cases represented 15.8% of the sample, with an imbalance ratio of 1:5.3. The child's sex is the one recorded in the survey (male or female), without distinction of gender. To avoid results inflated by using data that reveal the outcome (data leakage) [30], direct height in centimeters (HC3), raw weight (HC2) and any therapeutic intervention or clinical diagnosis recorded after the anthropometric measurement were excluded from the predictors. Unlike the official indicator [31], the sample was not restricted to children who slept in the household the night before (HV103); applying that filter changes the weighted prevalence from 16.2% to 16.1%.

$$Y = \begin{cases} 1, & \text{if HAZ} < -2.00 \text{ SD}, \\ 0, & \text{if HAZ} \geq -2.00 \text{ SD}. \end{cases} \tag{1}$$

The 45 final predictor variables are grouped into five domains: the child, the mother with her pregnancy and breastfeeding, the household and territory, water and sanitation, and housing (Table 1). These domains correspond to the immediate, underlying and basic causes of maternal and child undernutrition [1, 2]. The inclusion of altitude and adjusted hemoglobin is supported by their relationship with anemia in the Peruvian child population [28], and that of time to the water source and housing conditions by their appearance among the main predictors in DHS data [15].

2.2. Data preparation

Before variable selection, the codes of the different survey rounds were harmonized, missing values were handled, and categorical variables were encoded to obtain variables comparable across years. The biological consistency between the child and mother records, described in the previous subsection, was also verified.

Special survey codes (for example, "don't know", "not applicable" or "test not performed") were replaced with missing

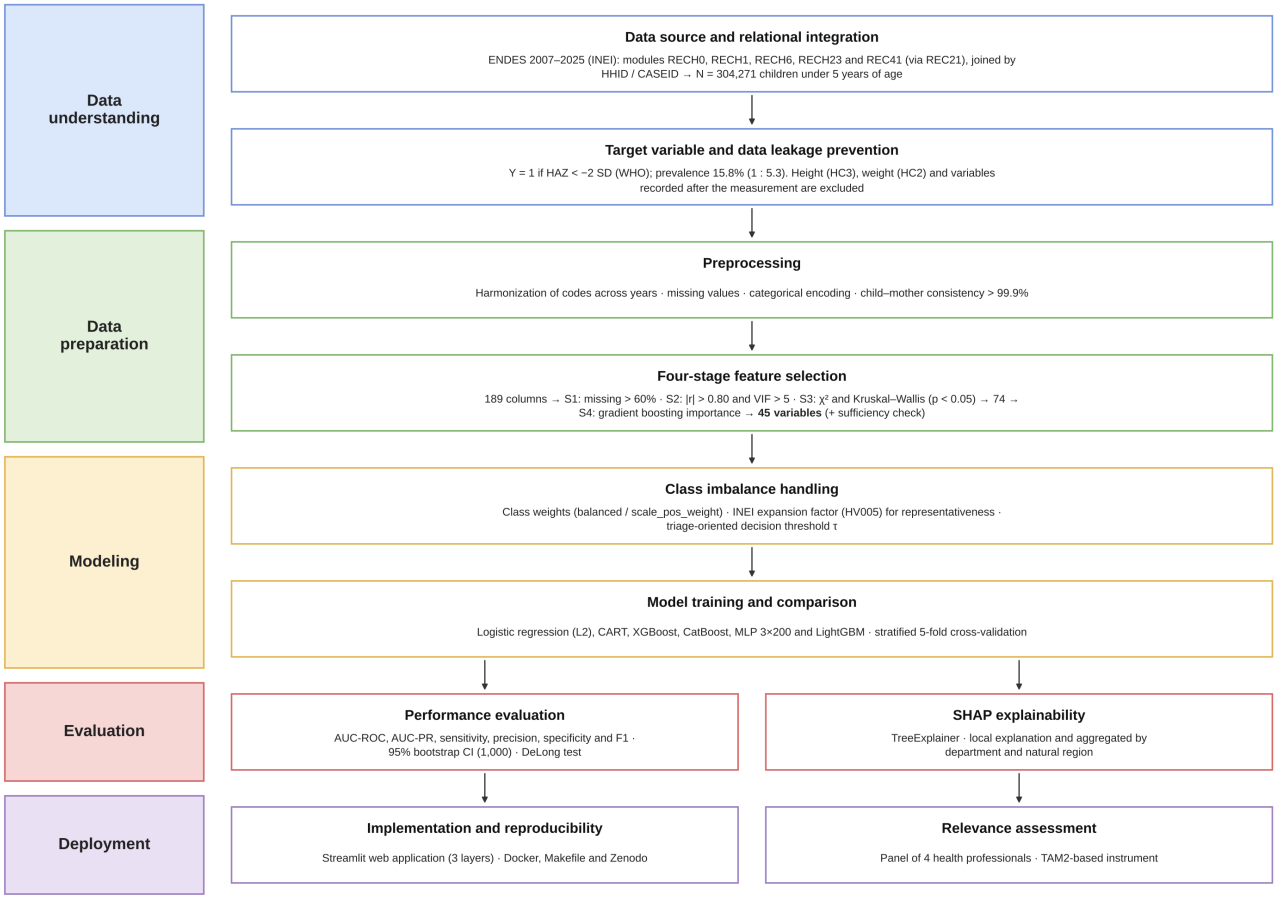


Figure 1: Methodological workflow of the study according to the CRISP-DM phases.

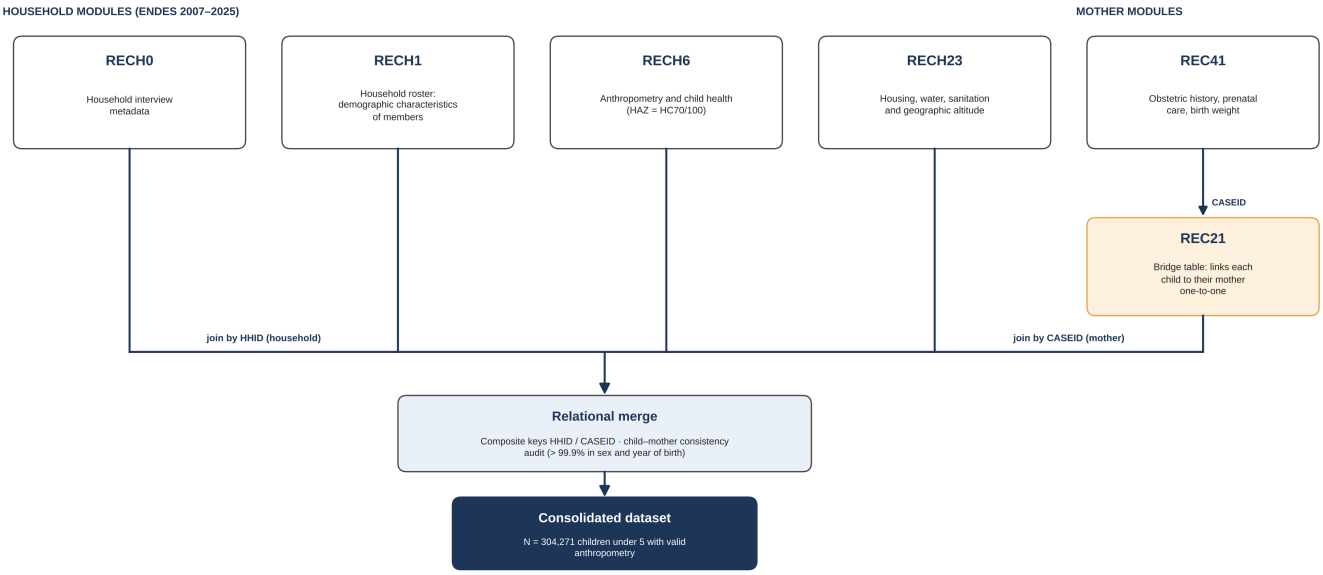


Figure 2: Relational integration of the 2007–2025 ENDES modules.

values, and categorical variables were encoded with the official INEI labels. Missing values in categorical variables were grouped into an "Unknown" category. Missing numerical values were imputed with the median, and continuous variables

Table 1: Predictor variables of the model grouped by domain (ENDES code in parentheses).

Domain (n)	Variables
Child (7)	Sex (HC27), Age in months (HC1), Birth weight (m19), Size at birth reported by the mother (M18), Altitude-adjusted hemoglobin (HC56), Interval since the previous birth (HC63), Usual resident (HV102)
Mother, pregnancy and breastfeeding (11)	Mother's education (HC62), Prenatal visits (M14), Month of first prenatal visit (M13), Days of iron during pregnancy (M46), Antiparasitic drugs during pregnancy (M60), Place of delivery (m15), Cesarean delivery (M17), Postpartum vitamin A (M54), Hours until breastfeeding initiation (m34), Formula in the first days (M55G), Duration of breastfeeding (M4)
Household and territory (8)	Wealth index (HV271), Area of residence (HV025), Altitude (HV040), Household members (HV012), Children under five in the household (HV014), Age of household head (HV220), Sex of household head (HV219), Traditional transport (SH77F)
Water and sanitation (10)	Water all day (SH42), Time to water source (HV204), Water treatment (HV237), Boiled water (HV237A), Bottled water (HV237G), Water stored in container (SH48), Container type (SH49), Container with lid (SH50), Shared toilet facility (HV225), Iodine in salt (HV234)
Housing (9)	Rooms (SH71), Bedrooms (HV216), Lighting source (SH70), Kitchen smoke outlet (HV240), Kitchen in separate room (HV242), Other cooking fuel (SH63), Glass windows (SH76B), Wooden windows (SH76C), Windows with screens (SH76D)

were standardized for the models that require it (logistic regression, XGBoost, decision tree and neural network), whereas LightGBM and CatBoost handle them natively.

With the preprocessed data, feature selection was carried out in four quantitative stages starting from the 189 raw columns. First, attributes with more than 60% of historical structural missing values were discarded. Second, repeated or highly correlated variables were removed using bivariate Pearson and Spearman correlations ($r > 0.80$) and variance inflation factors (VIF > 5). Third, chi-square tests of independence were applied to categorical variables and non-parametric Kruskal–Wallis tests to continuous variables; 74 statistically significant factors were retained ($p < 0.05$). Fourth, multivariate importance was assessed with gradient-boosted decision trees, and 45 predictor variables were selected.

To rule out that the selection had removed useful information, the effect of adding back the excluded variables was checked. Of the 78 columns available in the preprocessed dataset, 33 were not among the 45 selected. These were ranked by their relationship with stunting: for categorical attributes, the range between the highest and lowest prevalence among categories with at least 500 observations was used, whereas for continuous variables the statistic $2|\mathrm{AUC} - 0.5|$ was used.

Four experimental configurations were compared, the 45-variable model and three that added the top-ranked variables, keeping the partition, the weights, the deterministic seed and the classifier hyperparameters constant. The results of this check are presented in Section 3.2.

2.3. Modeling

Before training the models, class imbalance was addressed, since for every stunted child there are 5.3 children without stunting. So that the models would not ignore the minority class, class weights were used: misclassifying a stunted child costs more than misclassifying a non-stunted one, in inverse proportion to the frequency of each class. They were applied as balanced class weights in logistic regression, the decision tree, LightGBM and CatBoost, and as `scale_pos_weight` in XGBoost; the scikit-learn neural network does not support weights and was trained without them.

No synthetic records were generated (for example, with SMOTE). The minority class already has more than 48,000 real cases, synthetic records would not have their own expansion factor, and, without them, the metrics are computed on the real distribution of the population.

Separately, the INEI expansion factor, $w_i = \mathrm{HV005}_i/1{,}000{,}000$, was applied in training and in the computation of the metrics. This factor does not correct the imbalance: it corrects the representativeness of the survey, which oversamples some areas, so that the results reflect the country's population. Omitting it can overestimate the performance of models trained on complex survey data [32].

With this treatment, six supervised algorithms were evaluated: Logistic Regression with L2 regularization, CART Decision Tree, XGBoost, CatBoost, Multilayer Perceptron Neural Network (MLP with three hidden layers of 200 neurons each) and LightGBM; the three boosted tree ensembles are described in their original papers [33, 34, 35]. The comparison was performed on the 74 significant factors using stratified five-fold cross-validation and a deterministic seed (`random_state = 42`). LightGBM can group high-cardinality categorical variables through histograms, without resorting to one-hot encoding [36]. The choice of these algorithms, in particular tree-based ensembles, is consistent with studies on DHS data that report their usefulness for predicting child undernutrition [15, 14, 16]. Hyperparameters were set in advance, without a systematic search (Table 2); those not listed keep the default values of each library.

Finally, the decision threshold τ was set with a preventive triage criterion, which prioritizes detecting positive cases, instead of assuming the conventional value of 0.5, since a high AUC-ROC is not enough to use a model in practice without choosing the threshold according to its purpose [37].

Table 2: Configuration of the algorithms compared (`random_state = 42` in all).

Model	Configuration
Logistic regression	L2 regularization; balanced class weights; maximum of 500 iterations
Decision tree	Maximum depth of 5; balanced class weights
XGBoost	300 trees; `scale_pos_weight` equal to the ratio of negative to positive class; `min_child_weight` = 10^{-7}
CatBoost	300 iterations; balanced class weights; native categorical variables
Neural network	Three hidden layers of 200 neurons; up to 50 iterations with early stopping; no weighting
LightGBM	300 trees; balanced class weights; native categorical variables

2.4. Evaluation

Performance evaluation was based on the out-of-fold probabilities of the 304,271 records, following the recommendations for reporting prediction models in health [38]. AUC-ROC, sensitivity, precision and specificity were computed, all weighted with the expansion factor, since unweighted metrics do not represent performance in the population [39]. The uncertainty of the estimates was quantified with 1,000 non-parametric bootstrap replicates [40] and their 95% confidence intervals. The selected model was compared with the reference models using DeLong's test for correlated ROC curves [41]. Because this test is rank-based and does not support sampling weights, it was applied to the unexpanded sample, using the same individuals for all models.

In addition to measuring performance, the contribution of each variable to the predictions was computed with SHAP (SHapley Additive exPlanations; TreeExplainer algorithm) [42]. This approach decomposes the predicted probability into the sum of the population base value and the marginal contributions of each of the 45 predictor variables (Equation 2). At the local level, the system separates the individual prediction into risk factors and protective conditions. At the global and territorial levels, SHAP values were aggregated over a representative sample of 10,000 records to build risk maps and differentiated determinant profiles for the 25 departments and the three natural regions of the country.

$$f(x) = \phi_0 + \sum_{j=1}^{M} \phi_j(x). \tag{2}$$

2.5. Deployment

The solution was implemented as an interactive web application in Streamlit, organized in three layers (Figure 3) and available at `https://stunting-risk-peru.streamlit.app`. The presentation layer includes the departmental choropleth view based on GeoJSON geometries, the assessment of a single child with the explanation of their score, and the assessment of a group of children from a CSV or Excel file, which returns a list sorted by risk. The logic layer contains the LightGBM inference engine, the priority threshold selector according to the chosen coverage, and the computation of each variable's contribution with SHAP values at run time. The data layer contains the serialized model and its metadata, the coverage–threshold curve, the SHAP values aggregated by department and the departmental geometries; the master dataset in Parquet format is used only in the training workflow.

Reproducibility was addressed according to the FAIR principles through the modularization of the code in the Python package `mnp/`, environment isolation with Docker (`python:3.12-slim`), workflow automation with Makefile, and the deposit of the code and the serialized model in Zenodo (code: `https://doi.org/10.5281/zenodo.23963410`; model: `https://doi.org/10.5281/zenodo.23962363`). The data are not redistributed: the workflow downloads them directly from the INEI portal. The workflow was run with Python 3.12 and the libraries LightGBM 4.6.0, XGBoost 3.2.0, CatBoost 1.2.10, scikit-learn 1.8.0 and SHAP 0.52.0.

Once the application was implemented, its relevance was assessed by a panel of four health professionals: three nutritionists and one nurse, all with at least ten years of clinical experience and a master's degree. The assessment was based on the Technology Acceptance Model [43] and its extension TAM2 [44], which is frequently adapted to health settings [45]; perceived usefulness was the central criterion, as in other evaluations of clinical decision support systems [46], because the purpose of the system is to provide actionable information to prioritize health care.

The choice of acceptance model followed the decision framework of Britto [47], which compares TAM, TAM2, TAM3, UTAUT and UTAUT2 according to the context, the usage regime, the adoption stage, the explanatory focus and the granularity. TAM2 was chosen because this was an organizational evaluation of initial acceptance with a small panel. From TAM3 [48], only perceptions of external control were incorporated, to determine whether the data required by the system are available in clinical practice; the image and voluntariness constructs were omitted, as they do not apply to specialists who evaluate the tool without any obligation to adopt it.

The items were adapted to the health setting from the scales of Venkatesh and Davis [44], keeping in each construct statements with distinct content and at least two items. Table 3 details the constructs, their source model and the number of items in each. The instrument consists of 19 statements on a five-point Likert scale, an overall rating from 1 to 10 and four open-ended questions. Item K2 specifically asks about the availability of hemoglobin data, the only predictor that requires a procedure on the child. Responses are summarized by the median and interquartile range per construct and the percentage of agreement per item; a median of 4 or higher is considered favorable.

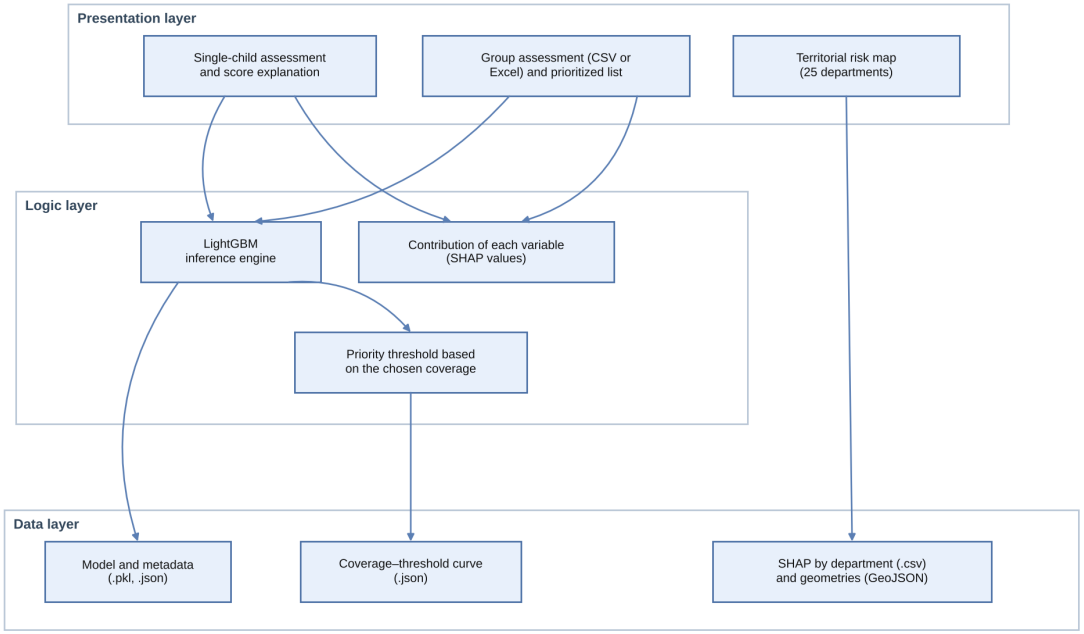


Figure 3: Three-layer architecture of the solution: presentation, logic and data.

Table 3: Constructs of the relevance assessment instrument, source model and items.

Construct	Source model	Items in the source	Items in the instrument
Job relevance	TAM2	2	2 (R1–R2)
Output quality	TAM2	2	2 (C1–C2)
Result demonstrability	TAM2	4	2 (D1–D2)
Subjective norm	TAM2	2	2 (S1–S2)
Perceived usefulness	TAM	4	3 (U1–U3)
Perceived ease of use	TAM	4	3 (F1–F3)
Intention to use	TAM	2	2 (I1–I2)
Perceptions of external control	TAM3	—	3 (K1–K3)

2.6. Ethical considerations

The ENDES microdata are publicly available, without restrictions on use, and contain no information that identifies the participants. The project, the assessment instrument, the final dataset and the application were reviewed by the Ethics Committee of Universidad Peruana Unión, which issued a favorable opinion (code XXXX). The panel assessment involved no intervention on children or their families, and its results were not used to make decisions about individuals. The specialists participated voluntarily, and their responses were analyzed in aggregate form, without individual identification.

3. Results

3.1. Sample characteristics

The sample included 304,271 children under five years of age, 51.0% boys and 65.6% urban residents. Prevalences were weighted with the survey expansion factor, as in the official indicator [31], so that each child counts according to the number of children they represent and the results estimate the situation of the country. Over the 19 years as a whole, the prevalence of stunting was 16.2% (Table 4). It was higher in rural areas (30.7%) than in urban areas (9.4%), in the Highlands (26.2%) and the Rainforest (21.8%) than in the Coast outside Lima (10.0%) and in Metropolitan Lima (5.4%), and between 12 and 23 months of age (19.8%).

By year, the weighted national prevalence fell from 29.2% in 2007 to 13.0% in 2017 and, since 2018, has remained between 11.3% and 12.4%, with 12.0% in 2025 (Figure 4). The gap between areas persists: in 2025, prevalence was 21.5% in rural areas and 7.9% in urban areas.

Table 4: Sample characteristics and weighted prevalence of stunting, 2007–2025.

Characteristic	*n*	% of sample	Prevalence (%)
Total	304,271	100.0	16.2
Sex			
Boys	155,149	51.0	17.1
Girls	149,122	49.0	15.2
Area of residence			
Urban	199,571	65.6	9.4
Rural	104,700	34.4	30.7
Natural region			
Metropolitan Lima	32,004	10.5	5.4
Rest of the Coast	85,227	28.0	10.0
Highlands	104,518	34.4	26.2
Rainforest	82,522	27.1	21.8
Age (months)			
0–5	23,615	7.8	11.2
6–11	30,742	10.1	12.5
12–23	60,560	19.9	19.8
24–59	189,354	62.2	16.2

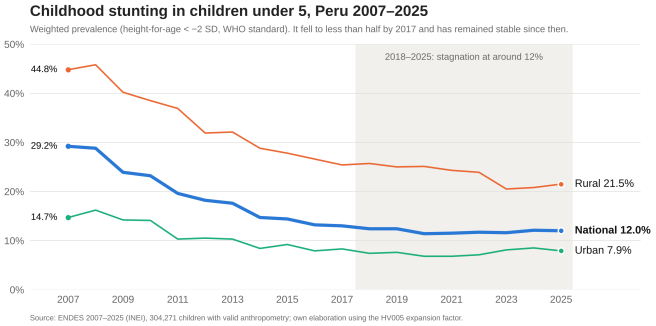


Figure 4: Weighted prevalence of stunting in children under five, Peru, 2007–2025.

3.2. Variable selection

The selection sequence reduced the 189 raw columns to 74 statistically significant factors and, finally, to the 45 predictor variables in Table 1.

In the check, the ranking identified ten attributes with a marked relationship with stunting: predominant material of the walls (HV214), floor (HV213) and roof (HV215); source of drinking water (HV201); department of residence (HV024); residual chlorine concentration in water (SH227); cooking fuel (HV226); and type of toilet facility (HV205), among others. These attributes make up the block of water, sanitation, hygiene and material housing conditions, recognized as an underlying determinant of linear growth faltering [1, 2] and found among the main predictors in studies based on DHS data [15]. None of them constitutes information leakage, as they describe structural household conditions observable before a home visit.

Adding the ten attributes did not improve any of the three configurations tested (water, sanitation and housing; territory; and both): the AUC-ROC (0.8295), sensitivity (76.44%) and precision (36.36%) were identical to those of the 45-variable model, because the model did not use the added variables. These attributes do contain information: a classifier trained only with the water, sanitation and housing block achieved an AUC-ROC of 0.6895. Nor do they duplicate the wealth index, as it explained between $\eta^2 = 0.038$ and $\eta^2 = 0.098$ of the variance of each one. The 45-variable subset already represents water, sanitation and housing conditions through nineteen correlated attributes, so the exclusion of the candidate variables is a methodologically sound decision.

3.3. Comparative performance of the models

Across the five validation folds, with the threshold of 0.5 (Table 5), LightGBM achieved the highest AUC-ROC (0.8297), followed by CatBoost (0.8283) and the neural network (0.8227); logistic regression, XGBoost and the decision tree ranged between 0.7800 and 0.7936. CatBoost obtained a slightly higher AUC-PR (0.5065 versus 0.4985), but with the threshold of 0.5 it detected only 29.1% of cases, and the neural network, trained without weights, 24.5%, compared with 76.4% for LightGBM, which also obtained the highest F1 (49.3%). LightGBM was selected for its discrimination and sensitivity. Retrained with the 45 final variables, it kept the same performance (AUC-ROC of 0.8295, sensitivity of 76.44% and precision of 36.36%).

3.4. ROC curves and confusion matrix

Each ROC curve shows, for all possible thresholds, the proportion of cases detected against the proportion of false positives; the diagonal corresponds to random classification, and the closer the curve is to the upper left corner, the better the model discriminates. The curves of LightGBM, CatBoost and the neural network overlap over most of their range, above those of logistic regression, XGBoost and the decision tree (Figure 5). In the fifth fold and with the threshold of 0.5, LightGBM correctly classified 75.3% of stunted children and 73.8% of children without stunting (Figure 6).

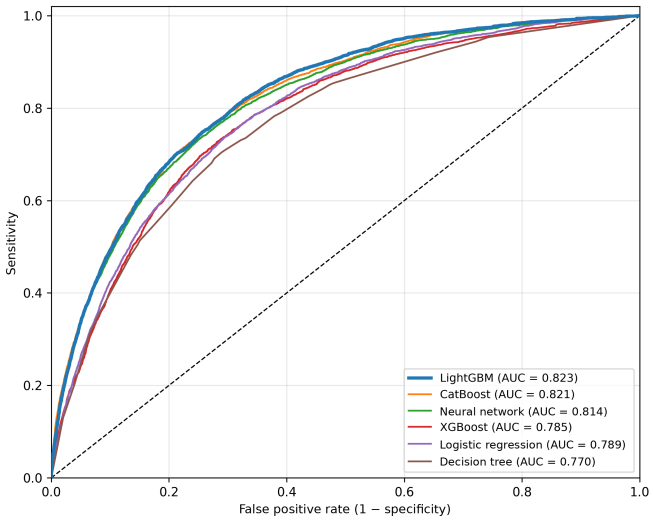


Figure 5: ROC curves of the six algorithms in the fifth validation fold.

Table 5: Performance of the six algorithms: mean (standard deviation) across the five folds, threshold of 0.5. Best value in bold.

Model	AUC-ROC	AUC-PR	Sensitivity (%)	Precision (%)	Specificity (%)	F1 (%)
Logistic regression	0.7925 (0.0024)	0.4173 (0.0050)	71.4 (0.7)	33.8 (0.3)	73.0 (0.4)	45.9 (0.2)
LightGBM	**0.8297 (0.0055)**	0.4985 (0.0057)	**76.4 (1.1)**	36.4 (0.3)	74.3 (0.4)	**49.3 (0.4)**
XGBoost	0.7936 (0.0052)	0.4249 (0.0067)	75.9 (0.9)	32.4 (0.3)	69.5 (0.5)	45.4 (0.3)
CatBoost	0.8283 (0.0057)	**0.5065 (0.0069)**	29.1 (0.9)	61.0 (0.7)	96.4 (0.1)	39.4 (1.0)
Decision tree	0.7800 (0.0067)	0.3914 (0.0050)	71.7 (1.5)	32.5 (0.5)	71.3 (1.0)	44.8 (0.3)
Neural network	0.8227 (0.0059)	0.4874 (0.0058)	24.5 (2.1)	**61.3 (1.4)**	**97.0 (0.4)**	35.0 (1.8)

Table 6: LightGBM metrics with 95% bootstrap confidence intervals.

Metric	Estimate	95% CI
AUC-ROC	0.8294	0.8267–0.8324
Sensitivity	76.43%	75.88–76.98%
Precision	36.36%	35.94–36.71%
F1	49.28%	48.85–49.66%

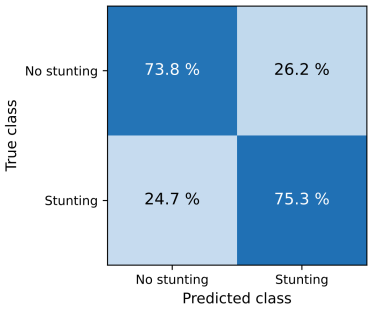


Figure 6: Confusion matrix of LightGBM in the fifth fold (threshold of 0.5).

3.5. Statistical reliability of the selected model

With 1,000 bootstrap replicates on the out-of-fold probabilities of the 304,271 records, the selected model achieved an AUC-ROC of 0.8294 (95% CI: 0.8267–0.8324), a sensitivity of 76.43% and a precision of 36.36% with the threshold of 0.5 (Table 6). With the threshold of 0.45, used in the application, sensitivity rises to 80.46% and precision falls to 34.04%. DeLong's test confirmed the superiority of LightGBM over XGBoost ($z = 62.32$) and logistic regression ($z = 61.48$), with $p < 10^{-300}$ in both cases and unweighted AUCs of 0.8193, 0.7775 and 0.7767, respectively. When hemoglobin, the only predictor that requires a procedure on the child, was excluded, the AUC-ROC decreased from 0.8294 to 0.8281 (difference of −0.0014; 95% CI: −0.0019 to −0.0009) and sensitivity from 76.43% to 76.30%.

3.6. Explainability

Global importance, which measures how much the model output changes on average for each variable without indicating whether it raises or lowers it, was computed on a random sample of 10,000 children and is concentrated in a few factors (Figure 7): household poverty (0.42), birth weight (0.41), altitude of residence (0.26) and size at birth (0.25) stand out clearly, and the top ten variables account for 68.0% of total importance.

At the local level, Figure 8 decomposes the model output for a synthetic high-risk profile (0.93); each bar is the contribution of one variable, and bars to the right raise the result: a birth weight of 2.4 kg, the lowest socioeconomic level, an age of 18 months and small size at birth are the factors that raise the result the most. These values describe associations learned from the survey, not causal relationships.

The weight of each factor varies across territories (Figure 9); in that figure, each row is a department and each column a factor, and the color indicates its relative weight compared with the other departments, so that darker shades show where that factor weighs more: in Puno, altitude, household poverty and low birth weight stand out, whereas in Huancavelica low birth weight, the birth interval and the age of the household head stand out.

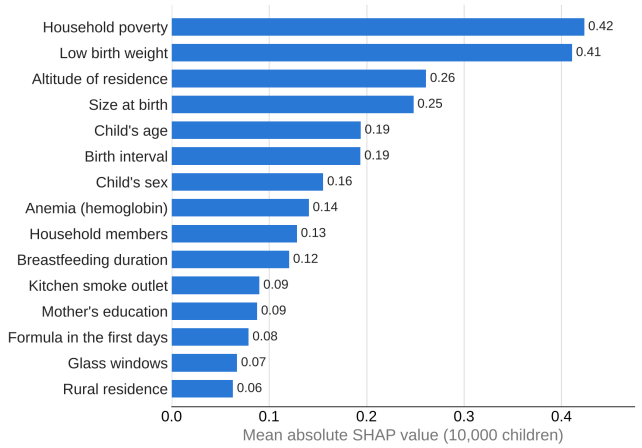


Figure 7: Global importance of the top 15 variables according to the mean absolute SHAP value.

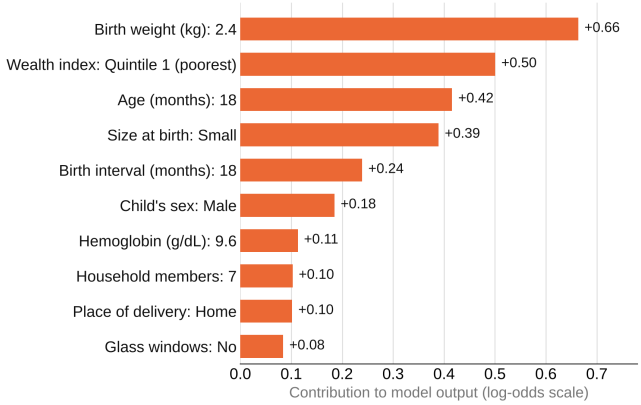


Figure 8: Local explanation of a synthetic high-risk profile (model output of 0.93).

3.7. Deployed system

The application makes it possible to assess a single child, with the explanation of the factors that raise or lower their score, and a group of children from a file, with a list sorted by risk (Figure 10). In the example profile, the score of 93 exceeds the threshold of 45 and the child is flagged as high priority (panel a); panel b shows which data raised that score. In a sample of 300 children from the 2025 ENDES, the application recognized the 45 columns and flagged 100 as high priority (panel c). Panel d corresponds to the territorial map: when a factor is chosen, it colors each department according to its weight and, when a department is chosen, it lists its factors with the greatest weight (Figure 9).

3.8. Relevance assessment by the panel

Four specialists completed the form (three nutritionists and one nurse; Coast and Rainforest; between 10 and 20 years of clinical experience). All constructs reached a median of 4 or higher (Table 7), and the overall rating had a median of 9 out of 10. The items with the lowest agreement were confidence that the assigned risk levels are correct and the availability in records of the requested data and of hemoglobin, with one evaluator scoring 3 in each case (75% agreement). The constructs closest to the model's explanations, output quality and result demonstrability, obtained 88% and 100% agreement, and no evaluator objected to the factors the system shows by child and by department (Figures 8 and 9); their suggestions focused on adding variables. With four evaluators, neither reliability nor the relationships between constructs were estimated: the results describe the panel's perception and cannot be generalized.

The open-ended responses converged on four themes: how current the data are and the equivalence of variables across years, raised by all four evaluators; extension to overweight and micronutrient deficiencies; greater territorial detail, by city or health facility, with feedback from local data; and the addition of further variables, such as exclusive breastfeeding, food consumption and chronic diseases.

Table 7: Panel ratings by construct (scale of 1 to 5). IQR: interquartile range.

Construct	Median	IQR	Agreement
Job relevance	4	4–5	100%
Output quality	4	4–4	88%
Result demonstrability	4.5	4–5	100%
Perceived usefulness	4	4–4.25	100%
Perceived ease of use	4	4–5	100%
Perceptions of external control	4.5	4–5	83%
Subjective norm	4.5	4–5	100%
Intention to use	4.5	4–5	100%
Overall rating (1 to 10)	9	8–10	—

4. Discussion

The selected model achieved an AUC-ROC of 0.83 on 304,271 children and nineteen years of survey data, a performance comparable to that of single-year DHS studies and higher than the pooled accuracy of 68.9% reported in a meta-analysis of those studies [26]. In Ethiopia, Bitew et al. [15] also obtained the best result with a boosted tree ensemble, and in Zambia Chilyabanyama et al. [16] achieved an accuracy of 79%. Talukder and Ahammed [14] reported a sensitivity of 94.7% in Bangladesh, but with a Cohen's kappa of 0.24, which shows that high sensitivity can hide low agreement. For this reason, this study reported sensitivity, precision, specificity, F1 and AUC-PR together, and not only accuracy, which, with a prevalence of 16%, favors models that flag almost no cases. This was the case for CatBoost and the neural network: with AUC-ROCs close to that of LightGBM, they detected fewer than 30% of stunted children with the threshold of 0.5. For the same reason, accuracies close to 100% obtained with synthetic oversampling [25] should be interpreted with caution.

With the threshold of 0.5, a precision of 36% means that, of every three children flagged, approximately one is stunted. For a triage tool this balance is acceptable, because the cost of a false positive is an additional assessment, whereas a false negative leaves a child at risk without care. On this basis, the threshold of 0.45 was set, which raises sensitivity to 80.5% at the cost of a precision of 34.0%.

The factors with the greatest weight match the framework of causes of maternal and child undernutrition [1, 2]: household poverty as a basic cause, birth weight and size at birth as a reflection of nutrition during pregnancy, and altitude, associated in Peru with hemoglobin concentration [28] and, in the Andes, with lower birth weight and length [49]. The importance of birth weight is consistent with growth faltering beginning in the first months of life [5]. The higher prevalence between 12 and 23 months matches the age pattern observed in 94 low- and middle-income countries [50], and the importance of poverty agrees with other studies using explainability [24] and with Colombia, where household wealth explains more than half of the rural–urban gap [51]. The variation in the weight of each factor across departments, such as altitude in Puno or the birth interval in Huancavelica, suggests that in-

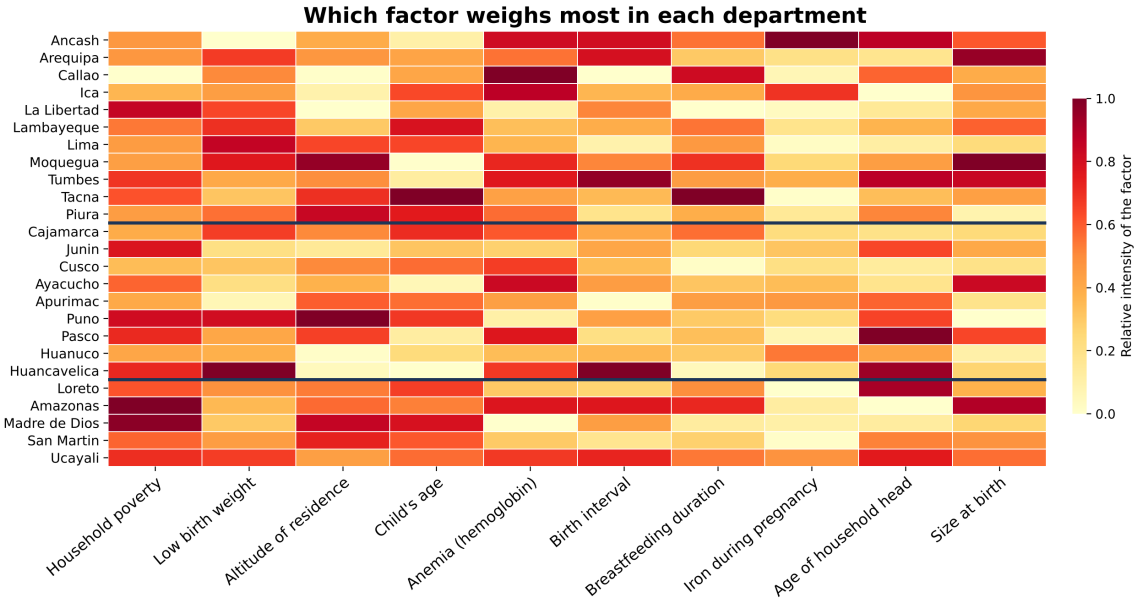


Figure 9: Relative weight of the top ten factors by department (0 = lowest; 1 = highest).

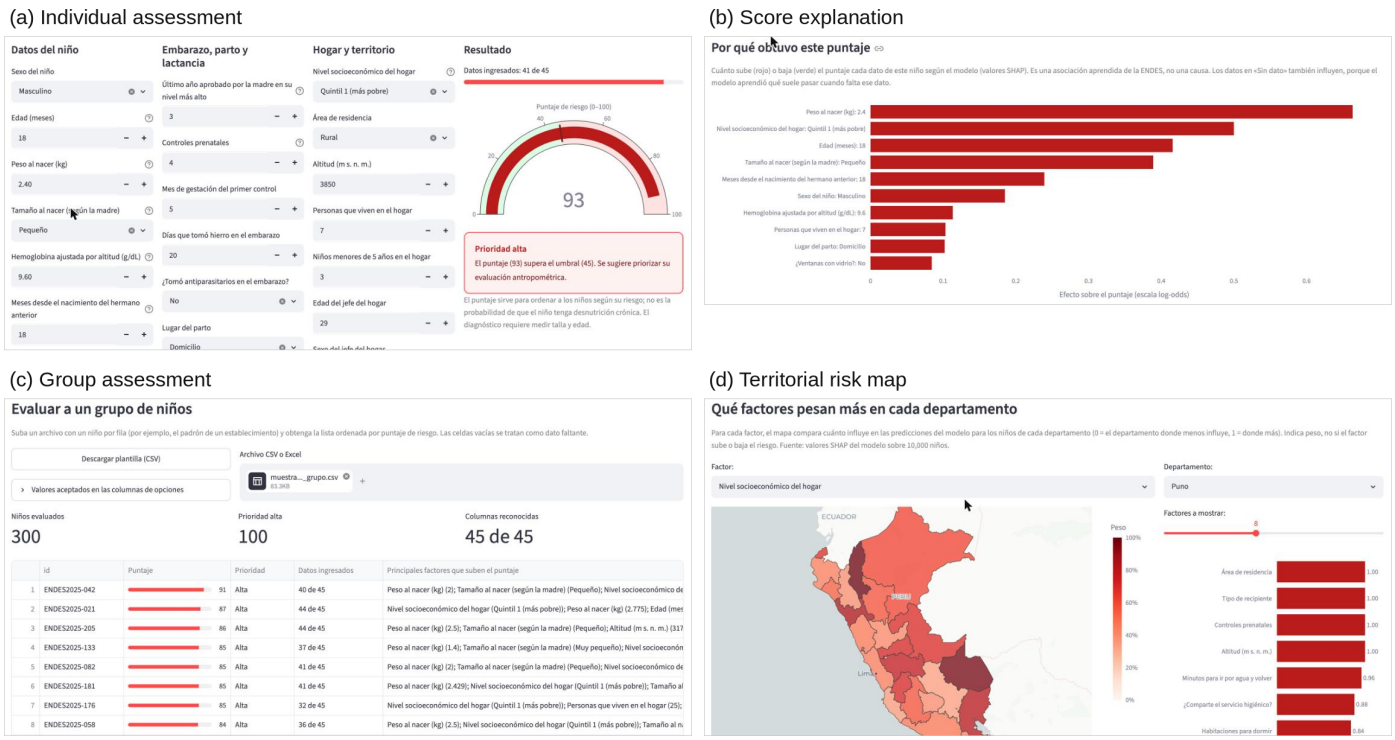


Figure 10: Application screens: (a) individual assessment, (b) explanation of the score, (c) assessment of a group of 300 children from the 2025 ENDES and (d) territorial risk map. The lines on the map delimit the study areas and do not necessarily represent recognized borders.

terventions could be prioritized according to the profile of each territory and not only according to prevalence.

The stagnation of prevalence at around 12% since 2018 and the rural gap of 21.5% versus 7.9% in urban areas match the territorial inequality described by Santos et al. [10] and a period in which the factors that drove the earlier reduction [9] seem to have exhausted their effect, and they reinforce the value of targeting resources.

The panel of specialists rated the usefulness and ease of use of the application favorably, and their comments focused on how current the data are and on adding new variables, not on the factors shown. This rating is similar to that of a machine

learning tool in primary care that 65% of its users considered useful [52], although actual adoption also depends on organizational factors that an initial evaluation does not measure [53].

This study has limitations. First, the ENDES is a cross-sectional survey, so the model identifies associations and does not allow causal inference. Second, cross-validation mixes the survey years; performance was not evaluated by training on one period and testing on another, nor with data external to the ENDES. Third, the wealth index (HV271) is computed in each round on its own scale, which may affect its comparability across years. Fourth, hyperparameters were set without a systematic search, and the comparison between algorithms was performed on the 74 significant factors. Fifth, hemoglobin requires a procedure on the child, although its exclusion reduced the AUC-ROC by only 0.0014. Sixth, the effect of weighting was not explicitly evaluated, although some studies indicate that the sampling design does not always change predictive performance [54]. Finally, a panel of four specialists does not allow the evaluation of the application to be generalized.

5. Conclusions

This study developed a model that predicts the risk of childhood stunting in Peru from the 2007–2025 ENDES microdata. Among six algorithms, LightGBM achieved the best balance between discrimination and case detection, with an AUC-ROC of 0.83 and a sensitivity of 80.5% with the triage threshold. Household poverty, birth weight, size at birth and altitude were the factors most associated with risk, with profiles that vary across departments.

The web application developed makes it possible to assess a single child or a group and explains each result, and it was rated favorably by health professionals as support for targeting and prioritizing care. Future work includes validating the model by survey period, correcting the scale of the wealth index across years, incorporating local data from health facilities, and extending the approach to overweight and micronutrient deficiencies.

Declaration of generative AI use in writing

During the preparation of this work, the authors used Claude (Anthropic) to improve the writing, readability and organization of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

[1] R. E. Black, C. G. Victora, S. P. Walker, Z. A. Bhutta, P. Christian, M. de Onis, et al., Maternal and child undernutrition and overweight in low-income and middle-income countries, The Lancet 382 (9890) (2013) 427–451. doi:10.1016/S0140-6736(13)60937-X.

[2] C. G. Victora, P. Christian, L. P. Vidaletti, G. Gatica-Domínguez, P. Menon, R. E. Black, Revisiting maternal and child undernutrition in low-income and middle-income countries: variable progress towards an unfinished agenda, The Lancet 397 (10282) (2021) 1388–1399. doi:10.1016/S0140-6736(21)00394-9.

[3] C. G. Victora, L. Adair, C. Fall, P. C. Hallal, R. Martorell, L. Richter, et al., Maternal and child undernutrition: consequences for adult health and human capital, The Lancet 371 (9609) (2008) 340–357. doi:10.1016/s0140-6736(07)61692-4.

[4] C. R. Sudfeld, D. Charles McCoy, G. Danaei, G. Fink, M. Ezzati, K. G. Andrews, et al., Linear growth and child development in low- and middle-income countries: A meta-analysis, Pediatrics 135 (5) (2015) e1266–e1275. doi:10.1542/peds.2014-3111.

[5] J. Benjamin-Chung, A. Mertens, J. M. Colford, A. E. Hubbard, M. J. van der Laan, J. Coyle, et al., Early-childhood linear growth faltering in low- and middle-income countries, Nature 621 (7979) (2023) 550–557. doi:10.1038/s41586-023-06418-5.

[6] M. de Onis, K. G. Dewey, E. Borghi, A. W. Onyango, M. Blössner, B. Daelmans, et al., The <scp>w</scp> orld <scp>h</scp> ealth <scp>o</scp> rganization's global target for reducing childhood stunting by 2025: rationale and proposed actions, Maternal amp; Child Nutrition 9 (S2) (2013) 6–26. doi:10.1111/mcn.12075.

[7] I. C. M. da Silva, G. V. França, A. J. Barros, A. Amouzou, J. Krasevec, C. G. Victora, Socioeconomic inequalities persist despite declining stunting prevalence in low- and middle-income countries, The Journal of Nutrition 148 (2) (2018) 254–258. doi:10.1093/jn/nxx050.

[8] M. Sobrino, C. Gutiérrez, A. J. Cunha, M. Dávila, J. Alarcón, Desnutrición infantil en menores de cinco años en Perú: tendencias y factores determinantes, Revista Panamericana de Salud Pública 35 (2) (2014) 104–112.

[9] L. Huicho, E. Vidal-Cárdenas, N. Akseer, S. Brar, K. Conway, M. Islam, et al., Drivers of stunting reduction in peru: a country case study, The American Journal of Clinical Nutrition 112 (2020) 816S–829S. doi:10.1093/ajcn/nqaa164.

[10] M. P. Santos, B. Turner, M. P. Chaparro, The double burden of malnutrition in Peru: an update with a focus on social inequities, The American Journal of Clinical Nutrition 113 (4) (2021) 865–873. doi:10.1093/ajcn/nqaa387.

[11] Z. A. Bhutta, J. K. Das, A. Rizvi, M. F. Gaffey, N. Walker, S. Horton, et al., Evidence-based interventions for improvement of maternal and child nutrition: what can be

done and at what cost?, The Lancet 382 (9890) (2013) 452–477. doi:10.1016/s0140-6736(13)60996-4.

[12] N. Scott, D. Delport, S. Hainsworth, R. Pearson, C. Morgan, S. Huang, et al., Ending malnutrition in all its forms requires scaling up proven nutrition interventions and much more: a 129-country analysis, BMC Medicine 18 (1) (Nov. 2020). doi:10.1186/s12916-020-01786-5.

[13] K. Curi-Quinto, E. Ortiz-Panozo, D. López de Romaña, Malnutrition in all its forms and socio-economic disparities in children under 5 years of age and women of reproductive age in peru, Public Health Nutrition 23 (S1) (2019) s89–s100. doi:10.1017/s136898001900315x.

[14] A. Talukder, B. Ahammed, Machine learning algorithms for predicting malnutrition among under-five children in Bangladesh, Nutrition 78 (2020) 110861. doi:10.1016/j.nut.2020.110861.

[15] F. H. Bitew, C. S. Sparks, S. H. Nyarko, Machine learning algorithms for predicting undernutrition among under-five children in Ethiopia, Public Health Nutrition (2021) 1–12doi:10.1017/S1368980021004262.

[16] O. N. Chilyabanyama, R. Chilengi, M. Simuyandi, C. C. Chisenga, M. Chirwa, K. Hamusonde, et al., Performance of machine learning classifiers in classifying stunting among under-five children in Zambia, Children 9 (7) (2022) 1082. doi:10.3390/children9071082.

[17] H. M. Fenta, T. Zewotir, E. K. Muluneh, A machine learning classifier approach for identifying the determinants of under-five child undernutrition in ethiopian administrative zones, BMC Medical Informatics and Decision Making 21 (1) (Oct. 2021). doi:10.1186/s12911-021-01652-1.

[18] E. K. Anku, H. O. Duah, Predicting and identifying factors associated with undernutrition among children under five years in ghana using machine learning algorithms, PLOS ONE 19 (2) (2024) e0296625. doi:10.1371/journal.pone.0296625.

[19] S. Ndagijimana, I. Kabano, E. Masabo, J. M. Ntaganda, Predicting stunting status among under-5 children in Rwanda using neural network model: Evidence from 2020 Rwanda demographic and health survey, F1000Research 13 (2025) 128. doi:10.12688/f1000research.141458.2.

[20] M. N. A. Khan, R. M. Yunus, A hybrid ensemble approach to accelerate the classification accuracy for predicting malnutrition among under-five children in subsaharan african countries, Nutrition 108 (2023) 111947. doi:10.1016/j.nut.2022.111947.

[21] A. Hendy, S. M. F. Abdelaliem, H. M. Sultan, S. H. Alahmedi, R. K. Ibrahim, E. M. E. Abdelrazek, et al., Unlocking insights: Using machine learning to identify wasting and risk factors in egyptian children under 5, Nutrition 131 (2025) 112631. doi:10.1016/j.nut.2024.112631.

[22] P. K. Arya, K. Sur, T. Kundu, S. Dhote, S. K. Singh, Unveiling predictive factors for household-level stunting in india: A machine learning approach using nfhs-5 and satellite-driven data, Nutrition 132 (2025) 112674. doi:10.1016/j.nut.2024.112674.

[23] V. T. Siy Van, V. A. Antonio, C. P. Siguin, N. P. Gordoncillo, J. T. Sescon, C. C. Go, et al., Predicting undernutrition among elementary schoolchildren in the philippines using machine learning algorithms, Nutrition 96 (2022) 111571. doi:10.1016/j.nut.2021.111571.

[24] T. Z. Yehuala, M. N. Aweke, H. W. Abuhay, M. A. Limenih, N. D. Baykemagn, G. G. Alemu, et al., Applying machine learning algorithms and explainable ai to predict stunting and identify determinants among children under five in ethiopia, Discover Artificial Intelligence 6 (1) (Apr. 2026). doi:10.1007/s44163-026-01129-y.

[25] W. R. Mgomezulu, P. Thangata, B. Mkandawire, N. Amoah, Advancing predictive analytics in child malnutrition: Machine, ensemble and deep learning models with balanced class distribution for early detection of stunting and wasting, Human Nutrition amp; Metabolism 42 (2025) 200340. doi:10.1016/j.hnm.2025.200340.

[26] B. Rao, M. Rashid, M. G. Hasan, G. Thunga, Machine learning in predicting child malnutrition: A meta-analysis of demographic and health surveys data, International Journal of Environmental Research and Public Health 22 (3) (2025) 449. doi:10.3390/ijerph22030449.

[27] R. Wirth, J. Hipp, CRISP-DM: Towards a standard process model for data mining, in: Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining, 2000, pp. 29–39.

[28] R. A. Accinelli, J. A. Leon-Abarca, Age and altitude of residence determine anemia prevalence in Peruvian 6 to 35 months old children, PLOS ONE 15 (1) (2020) e0226846. doi:10.1371/journal.pone.0226846.

[29] WHO Multicentre Growth Reference Study Group, WHO Child Growth Standards: Length/height-for-age, weight-for-age, weight-for-length, weight-for-height and body mass index-for-age. Methods and development, World Health Organization, Geneva, 2006.

[30] S. Kaufman, S. Rosset, C. Perlich, Leakage in data mining: formulation, detection, and avoidance, in: Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2011, pp. 556–563. doi:10.1145/2020408.2020496.

[31] Instituto Nacional de Estadística e Informática, Perú: Indicadores de Resultados de los Programas Presupuestales, Primer Semestre 2023, INEI, Lima, 2023.

[32] N. MacNell, L. Feinstein, J. Wilkerson, P. M. Salo, S. A. Molsberry, M. B. Fessler, et al., Implementing machine

learning methods with complex survey data: Lessons learned on the impacts of accounting sampling weights in gradient boosting, PLOS ONE 18 (1) (2023) e0280387. doi:10.1371/journal.pone.0280387.

[33] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD '16, ACM, 2016, p. 785–794. doi:10.1145/2939672.2939785.

[34] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, A. Gulin, CatBoost: unbiased boosting with categorical features, in: Advances in Neural Information Processing Systems 31, 2018, pp. 6638–6648.

[35] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, et al., LightGBM: A highly efficient gradient boosting decision tree, in: Advances in Neural Information Processing Systems 30, 2017, pp. 3146–3154.

[36] C. Bentéjac, A. Csörgő, G. Martínez-Muñoz, A comparative analysis of gradient boosting algorithms, Artificial Intelligence Review 54 (3) (2021) 1937–1967. doi:10.1007/s10462-020-09896-5.

[37] J. Deng, M. E. Elghobashy, K. Zang, S. K. Patel, E. Guo, K. Heybati, So you've got a high auc, now what? an overview of important considerations when bringing machine-learning models from computer to bedside, Medical Decision Making 45 (6) (2025) 640–653. doi:10.1177/0272989x251343082.

[38] K. G. M. Moons, D. G. Altman, J. B. Reitsma, J. P. A. Ioannidis, P. Macaskill, E. W. Steyerberg, et al., Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): explanation and elaboration, Annals of Internal Medicine 162 (1) (2015) W1–W73. doi:10.7326/M14-0698.

[39] A. S. Wadekar, J. P. Reiter, Evaluating binary outcome classifiers estimated from survey data, Epidemiology 35 (6) (2024) 805–812. doi:10.1097/ede.0000000000001776.

[40] B. Efron, R. J. Tibshirani, An Introduction to the Bootstrap, Chapman and Hall, New York, 1993.

[41] E. R. DeLong, D. M. DeLong, D. L. Clarke-Pearson, Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach, Biometrics 44 (3) (1988) 837. doi:10.2307/2531595.

[42] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems 30 (NIPS 2017), 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[43] F. D. Davis, Perceived usefulness, perceived ease of use, and user acceptance of information technology, MIS Quarterly 13 (3) (1989) 319–340. doi:10.2307/249008.

[44] V. Venkatesh, F. D. Davis, A theoretical extension of the technology acceptance model: four longitudinal field studies, Management Science 46 (2) (2000) 186–204. doi:10.1287/mnsc.46.2.186.11926.

[45] E. D. Matemba, G. Li, I. C. W. Gogan, B. J. Maiseli, Technology acceptance model: recent developments, future directions, and proposal for hypothetical extensions, International Journal of Technology Intelligence and Planning 12 (4) (2020) 315. doi:10.1504/IJTIP.2020.109769.

[46] S.-C. Chien, C.-H. Chien, C.-Y. Chen, P.-H. Chien, C.-K. Hsu, H.-C. Yang, et al., Using technology acceptance model to explore physicians' perspectives of clinical decision support system alerts, BMJ Health amp; Care Informatics 32 (1) (2025) e101128. doi:10.1136/bmjhci-2024-101128.

[47] N. P. Britto, A decision framework for selecting technology acceptance and use models: TAM, TAM2, TAM3, UTAUT, and UTAUT2 (2026). arXiv:2608.08333.

[48] V. Venkatesh, H. Bala, Technology acceptance model 3 and a research agenda on interventions, Decision Sciences 39 (2) (2008) 273–315. doi:10.1111/j.1540-5915.2008.00192.x.

[49] J. I. Martínez, E. M. Román, E. L. Alfaro, C. Grandi, J. E. Dipierri, Geographic altitude and prevalence of underweight, stunting and wasting in newborns with the intergrowth-21st standard, Jornal de Pediatria 95 (3) (2019) 366–373. doi:10.1016/j.jped.2018.03.007.

[50] O. Karlsson, R. Kim, G. M. Moloney, A. Hasman, S. V. Subramanian, Patterns in child stunting by age: A cross-sectional study of 94 low- and middle-income countries, Maternal amp; Child Nutrition 19 (4) (2023). doi:10.1111/mcn.13537.

[51] E. Cardenas, A. M. Osorio, O. J. Barandica, S. M. Pico Fonseca, Mind the gap! socioeconomic determinants of the stunting urban-rural gap for children in colombia, Child Indicators Research 15 (2) (2022) 415–432. doi:10.1007/s12187-021-09880-7.

[52] W. E. Herter, J. Khuc, T. N. Bonten, R. A. Verheij, M. E. Numans, N. H. Chavannes, Usability and usefulness of machine learning–based clinical decision support software in primary care: Survey of users in a prospective observational study, JMIR Medical Informatics 14 (2026) e80527–e80527. doi:10.2196/80527.

[53] M. Ji, G. Z. Genchev, H. Huang, T. Xu, H. Lu, G. Yu, Evaluation framework for successful artificial intelligence–enabled clinical decision support systems: Mixed methods study, Journal of Medical Internet Research 23 (6) (2021) e25929. doi:10.2196/25929.

[54] Z. Si, J. Li, E. Leary, Machine learning in public health informatics: Evidence that complex sampling structures may not be needed for prediction models with imbalanced outcomes, Annals of Epidemiology 102 (2025) 75–80. doi:10.1016/j.annepidem.2024.12.016.